Tribhuvan University

Faculty of Science and Technology

OFFICE OF THE DEAN

2080

Bachelor of Science in Computer Science and Information Technology

Course Title: **Mathematics II**

Code No: **MTH168** Semester: **2nd Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. What is a system of linear equations? When the system is consistent? Find a condition on g, h, k that makes the system consistent$$x_1 - 4x_2 + 7x_3 = g$$$$3x_2 - 5x_3 = h$$$$-2x_1 + 5x_2 - 9x_3 = k.$$

2. Let $A = \begin{bmatrix} 1 & -5 & -7 \\ -3 & 7 & 5 \end{bmatrix}, u = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, b = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$. Define $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ by $T(x) = Ax$(a). Compute $T(u)$(b). Find a vector $x \in \mathbb{R}^3$ whose image under T is b.(c) Is x unique?

3. Find a least square solution of $Ax = b$ where $A = \begin{bmatrix} 1 & -3 & -3 \\ 1 & 5 & 1 \\ 1 & 7 & 2 \end{bmatrix}, b = \begin{bmatrix} 5 \\ -3 \\ -5 \end{bmatrix}$. Also, compute the associated least square error.

Section B

Attempt any Eight questions[8×5=40]

4. Are the vectors $v_1 = \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 10 \\ 2 \\ 1 \end{bmatrix}, v_3 = \begin{bmatrix} -5 \\ 0 \\ 6 \end{bmatrix}$ linearly independent? Justify.

5. Find LU factorization of $\begin{bmatrix} 2 & 3 & 4 \\ 4 & 5 & 10 \\ 4 & 8 & 2 \end{bmatrix}$

6. Compute $\text{Det } A$ where $A = \begin{bmatrix} 2 & -8 & 6 & 8 \\ 3 & -9 & 5 & 10 \\ -3 & 0 & 1 & -2 \\ 1 & -4 & 0 & 6 \end{bmatrix}$.

7. Show that $H = \{(a - 3b, b - a, a, b) : a, b \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^4$.

8. Is $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$ an eigenvector of $\begin{bmatrix} 5 & -3 \\ -4 & 9 \end{bmatrix}$? If so, find the eigenvalue.

9. Let $u = (1, -2, 2, 0)$. Find a unit vector v in the same direction as u.

10. Find the basis and dimension of $\text{Nul } A$ where $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 7 & 8 \end{bmatrix}$.

11. Define group. Show that $(\mathbb{Z},.)$ doesn't form a group.

12. Show that every field is an integral domain.